

Final Exam For Second Semester (2024/2025)



CS First Year students ONLY

Leader's :

1 - Jannah Ahmed

2 - Kareem Yousry

امتحان Final Exam for Second Semester (2024/2025)

Course Name: Discrete Mathematics	Course Code: BS103	Bylaw: 2024
Year: First	Model (A)	Program: Computer Sciences
Date of Exam: Sunday 25/5/2025	Time Allowed: 2 hours	Total Points: 60
Course Instructor: Assist. Prof. Dr/ Saeed Hemeda	No. of questions: (3)	No. of pages: (2)
Dean Approval:		

Exam instructions

- Make sure that the question form is identical to the answer sheet form
- The answer sheet must be in good condition from the moment received until it is delivered after taking the exam
- Do not bend the answer sheet or lean on it while answering

Answer the following questions:

Question One: [20 points]

(ILOs: a1, a2, a3, b1, b2, c1, c2, d1)

Choose the Correct answer:

- 1) If $A = \{1, 2, 3, 4\}$ and R is a relation on A such that: $R = \{(1, 2), (1, 3), (1, 4)\}$, then the converse of R is $R^T = \dots\dots\dots$

A) $\{(2, 3), (3, 1), (4, 2)\}$ B) $\{(2, 1), (3, 1), (4, 1)\}$ C) $\{(1, 1), (3, 3), (4, 4)\}$ D) $\{(1, 2), (1, 3), (1, 4)\}$

- 2) Let $S = \{v_0, v_1, v_2, \dots, v_n\}$ be the set of variables in a certain computer program. If $v_i = v_j$ means the values of v_i and v_j are the same, then at any moment during the course of running the program " $=$ " is $\dots\dots\dots$ relation on S .

A) an equivalence B) symmetric C) transitive D) one to one

- 3) If two edges of a graph are connected by a single vertex, they are called $\dots\dots\dots$

A) adjacent edges B) empty edges C) parallel edges D) finite edges

- 4) Suppose A and B are two non-empty sets then: $B + B^c B = \dots\dots\dots$

A) B^c B) B C) $\emptyset$ D) U

- 5) If p is true and q is false, then $(\sim P \vee q) \rightarrow p$ is $\dots\dots\dots$

A) false B) true C) q D) P

- ❖ If $A = \{0, 1, 2\}$, R and S are two relations on A where: $R = \{(0, 2), (1, 0), (1, 2), (2, 0)\}$ and $S = \{(1, 0), (1, 1), (1, 2), (2, 2)\}$, then;

- 6) $M_R = \dots\dots\dots$

A) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ B) $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ C) $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ D) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

- 7) $M_S = \dots\dots\dots$

A) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ B) $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ C) $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ D) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

❖ If R is a relation on $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (1, 1), (2, 4), (3, 1), (4, 3)\}$;

8) $R^c = \dots\dots$

- | | |
|---|---|
| A) $\{(1, 3), (1, 4), (2, 3), (2, 2), (2, 1), (4, 1)\}$ | B) $\{(1, 3), (1, 4), (2, 3), (2, 2), (2, 1), (4, 1), (4, 2), (4, 4), (3, 2), (3, 3), (3, 4)\}$ |
| C) $\{(1, 3), (4, 4), (2, 3), (2, 2), (2, 1), (1, 1)\}$ | D) $\{(2, 1), (1, 1), (4, 2), (1, 3), (3, 4)\}$ |

9) $M_{R^c} = \dots\dots$

- | | | | |
|---|---|---|---|
| A) $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$ | B) $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$ | C) $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$ | D) $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$ |
|---|---|---|---|

10) A graph that has finite number of vertices and edges is called $\dots\dots$

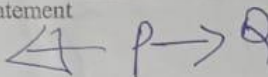
- | | | | |
|----------------|-------------------|-------------------|-----------------|
| A) empty graph | B) infinite graph | C) compound graph | D) finite graph |
|----------------|-------------------|-------------------|-----------------|

Question Two: [20 points]

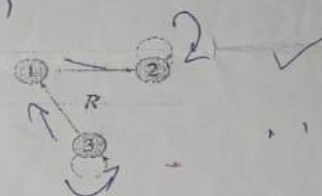
(ILOs: a1, a4, b1, b3, b4, c1, c2, d1, d2)

1) Deduce the result of the compound statement

$$[(K \rightarrow N) \wedge (N \rightarrow S)] \rightarrow (K \rightarrow S).$$



2) If $A = \{1, 2, 3\}$ and R is a relation on A defined by the following diagram:



Find the relation $(\overline{R^T} - R)$ and its matrix and graph its diagram.

Question Three: [20 points]

(ILOs: a2, a3, b1, b2, b4, c1, c3, d1, d2)

1) The relations $R = \{(a, a), (b, a), (b, d), (c, c), (c, d), (d, a), (d, c)\}$ and

$S = \{(a, b), (b, a), (c, a), (c, d), (d, a), (d, b)\}$ are defined on set $B = \{a, b, c, d\}$,

Find: $M_R \otimes M_S$ in matrix form.

2) Use mathematical induction to prove that: $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$, for all positive integers n .

End of Questions

Good Luck,,

Examiner:

Assist. Prof. Dr/ Saeed Hemeda

Assist. Prof. Hassan Algendy